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Builds systems from scratch across PL, ML, and the web.

EDUCATION

UC Irvine • B.S. in Computer Science w/ Honors, 3.9 / 4.0 GPA Sep 2024 – Jun 2026

- graduated in 2 years
- ICS Honors Program
- UROP Fellow
- Data Structures
- Algorithms
- Software Eng.
- Compilers
- Cryptography
- Formal Lang. & Automata
- Quantum Computation
- Distributed Systems

PROJECTS

manifold • Rust, wgpu, CEF, Servo Sep 2026 – Present

- Built a browser running each tab in its own engine process (Blink via CEF/WebView2, Servo, WebKitGTK) over IPC, with in-place engine hot-swap.
- Added crash recovery, journaled sessions, and a force-directed tab graph on **wgpu**.

scratchtape • Rust, wgpu, cubeccl Sep 2026 – Present

- Built an ML engine from scratch (autodiff, layers, GPU backend); every primitive is gradient-checked.
- Measured DX12 at $1.6\text{--}5.8\times$ lower overhead than Vulkan on small matmuls; up to 900 GFLOP/s.

tatic • Rust, WebAssembly Mar 2026 – Present

- Built a **JIT** caching compiled **WebAssembly** by content hash; fixed a leak found by benchmark.

cereal • C99 Sep 2026 – Present

- Wrote a C99 compiler front end in C99 that matches gcc-13 diagnostics on 96% of 3,700+ gcc.dg tests.

Apps & web • TypeScript, Svelte, Tauri Sep 2026 – Present

- **arrow-game**: strict-TS browser roguelite, no runtime dependencies; 300+ tests on Bun, Node.
- **lollipoppy**: Tauri (Rust) + Svelte desktop app that drives the League client.

SKILLS AND INTERESTS

Proficient in Rust, C / C++, JS / TS, Python, Svelte, Tailwind.

Familiar with Git, HTML / CSS, Nix, LaTeX, Typst, Claude Code, Unix systems & shells.

Acquainted with most mainstream languages, e.g., C#, Go, Haskell, Java, Lua, Ruby, SQL.

Domains: compilers & PL, ML & GPU compute, browsers, web & desktop apps, developer tooling.

EXPERIENCE

Learning Assistant • UC Irvine Mar 2025 – Jun 2026

- Delivered supplementary pedagogy for **Python** and upper-div **Algorithms**, e.g., Master Theorem.
- Spearheaded study sessions to identify and guide students through gaps in understanding.

Intern • LGBTQIA+ Identity Commission @ UC Irvine Oct 2024 – Jun 2025

- Led a team of 3 to organize and host themed Gayme Show™.

Programmer • Video Game Development Club @ UC Irvine Sep 2024 – May 2025

- Worked with team of 13 to create typing game in **C#** and **Unity**; led dev of the main UI window.